

Лабораторная работа №14

Администрирование локальных сетей

Ибрахим Мохсейн Алькамаль

2026-05-16

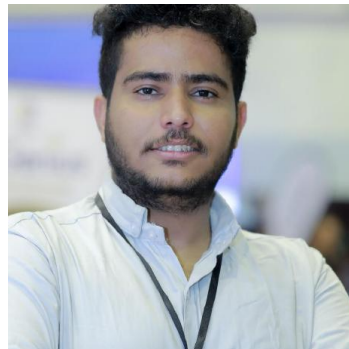
Содержание I

- 1 Информация
- 2 Цель работы
- 3 Выполнение лабораторной работы
- 4 Выводы

Раздел 1

Информация

- Ибрахим Мохсейн Алькамаль
- Российский университет дружбы народов
- GitHub



Раздел 2

Цель работы

Цель работы

- Настройка взаимодействия сетей через провайдера • Настройка статической маршрутизации между центральной сетью, q42 и Sochi

Раздел 3

Выполнение лабораторной работы

Настройка транковых соединений сети провайдера

- Настройка trunk-интерфейсов FastEthernet0/3 и FastEthernet0/4
- Создание VLAN 5 и VLAN 6
- Организация связи между филиалами и центральной сетью


```
provider-ealkamal-sw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
provider-ealkamal-sw-1(config)#interface f0/3
provider-ealkamal-sw-1(config-if)#switchport mode trunk

provider-ealkamal-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

provider-ealkamal-sw-1(config-if)#exit
provider-ealkamal-sw-1(config)#interface f0/4
provider-ealkamal-sw-1(config-if)#switchport mode trunk

provider-ealkamal-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

provider-ealkamal-sw-1(config-if)#exit
provider-ealkamal-sw-1(config)#vlan 5
provider-ealkamal-sw-1(config-vlan)#name q42
provider-ealkamal-sw-1(config-vlan)#exit
provider-ealkamal-sw-1(config)#interface vlan5
provider-ealkamal-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-ealkamal-sw-1(config-if)#no shutdown
provider-ealkamal-sw-1(config-if)#exit
provider-ealkamal-sw-1(config)#vlan 6
provider-ealkamal-sw-1(config-vlan)#name sochi
provider-ealkamal-sw-1(config-vlan)#exit
provider-ealkamal-sw-1(config)#interface vlan6
provider-ealkamal-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdown
provider-ealkamal-sw-1(config-if)#no shutdown
provider-ealkamal-sw-1(config-if)#exit
provider-ealkamal-sw-1(config)#
```

```
msk-donskaya-ealkamal-gw-1>enable
Password:
msk-donskaya-ealkamal-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-ealkamal-gw-1(config)#interface f0/1.5
msk-donskaya-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-ealkamal-gw-1(config-subif)#encapsulation dot1Q 5
msk-donskaya-ealkamal-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-ealkamal-gw-1(config-subif)#description q42
msk-donskaya-ealkamal-gw-1(config-subif)#exit
msk-donskaya-ealkamal-gw-1(config)#interface f0/1.6
msk-donskaya-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-ealkamal-gw-1(config-subif)#encapsulation dot1Q 6
msk-donskaya-ealkamal-gw-1(config-subif)#ip address 10.128.255.5 255.255.255.252
msk-donskaya-ealkamal-gw-1(config-subif)#description sochi
msk-donskaya-ealkamal-gw-1(config-subif)#exit
msk-donskaya-ealkamal-gw-1(config)#exit
msk-donskaya-ealkamal-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-ealkamal-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-ealkamal-gw-1#
```

Рис. 2: Настройка подинтерфейсов маршрутизатора центральной сети

```
msk-q42-ealkamal-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-ealkamal-gw-1(config)#interface f0/1
msk-q42-ealkamal-gw-1(config-if)#no shutdown

msk-q42-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-q42-ealkamal-gw-1(config-if)#exit
msk-q42-ealkamal-gw-1(config)#interface f0/1.5
msk-q42-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-q42-ealkamal-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-ealkamal-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
msk-q42-ealkamal-gw-1(config-subif)#description donskaya
msk-q42-ealkamal-gw-1(config-subif)#exit
msk-q42-ealkamal-gw-1(config)#exit
msk-q42-ealkamal-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-ealkamal-gw-1#write memory
Building configuration...
[OK]
msk-q42-ealkamal-gw-1#
```

Рис. 3: Настройка VLAN 5 на маршрутизаторе сети q42

Настройка коммутатора сети Sochi

```
sch-sochi-ealkamal-sw-1>enable
Password:
sch-sochi-ealkamal-sw-1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-ealkamal-sw-1(config)#interface f0/23
sch-sochi-ealkamal-sw-1(config-if)#switchport mode trunk
sch-sochi-ealkamal-sw-1(config-if)#exit
sch-sochi-ealkamal-sw-1(config)#interface f0/24
sch-sochi-ealkamal-sw-1(config-if)#switchport mode trunk
sch-sochi-ealkamal-sw-1(config-if)#exit
sch-sochi-ealkamal-sw-1(config)#vlan 6
sch-sochi-ealkamal-sw-1(config-vlan)#name sochi
sch-sochi-ealkamal-sw-1(config-vlan)#exit
sch-sochi-ealkamal-sw-1(config)#interface vlan6
sch-sochi-ealkamal-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

sch-sochi-ealkamal-sw-1(config-if)#no shutdown
sch-sochi-ealkamal-sw-1(config-if)#exit
sch-sochi-ealkamal-sw-1(config)#
```

Рис. 4: Настройка trunk-портов коммутатора сети Sochi

Настройка маршрутизатора сети Sochi

```
sch-sochi-ealkamal-gw-1>enable
Password:
sch-sochi-ealkamal-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-ealkamal-gw-1(config)#interface f0/0
sch-sochi-ealkamal-gw-1(config-if)#no shutdown

sch-sochi-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

sch-sochi-ealkamal-gw-1(config-if)#exit
sch-sochi-ealkamal-gw-1(config)#interface f0/0.6
sch-sochi-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

sch-sochi-ealkamal-gw-1(config-subif)#encapsulation dot1Q 6
sch-sochi-ealkamal-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
sch-sochi-ealkamal-gw-1(config-subif)#description donskaya
sch-sochi-ealkamal-gw-1(config-subif)#exit
sch-sochi-ealkamal-gw-1(config)#exit
sch-sochi-ealkamal-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-ealkamal-gw-1#|
```

Рис. 5: Настройка подинтерфейса VLAN 6 на маршрутизаторе сети Sochi

Настройка VLAN сети q42

• Создание подинтерфейсов FastEthernet0/0.201 и FastEthernet1/0.202

```
msk-q42-ealkamal-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-ealkamal-gw-1(config)#interface f0/0
msk-q42-ealkamal-gw-1(config-if)#no shutdown

msk-q42-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-q42-ealkamal-gw-1(config-if)#exit
msk-q42-ealkamal-gw-1(config)#interface f0/0.201
msk-q42-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-ealkamal-gw-1(config-subif)#encapsulation dot1Q 201
msk-q42-ealkamal-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
msk-q42-ealkamal-gw-1(config-subif)#description q42main
msk-q42-ealkamal-gw-1(config-subif)#exit
msk-q42-ealkamal-gw-1(config)#interface f1/0
msk-q42-ealkamal-gw-1(config-if)#no shutdown

msk-q42-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

msk-q42-ealkamal-gw-1(config-if)#exit
msk-q42-ealkamal-gw-1(config)#interface f1/0.202
msk-q42-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-ealkamal-gw-1(config-subif)#encapsulation dot1Q 202
msk-q42-ealkamal-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
msk-q42-ealkamal-gw-1(config-subif)#description q42management
msk-q42-ealkamal-gw-1(config-subif)#exit
msk-q42-ealkamal-gw-1(config)#exit
msk-q42-ealkamal-gw-1#
%SYS-5-CONFIC-1: Configured from console by console
```

```
msk-hostel-ealkamal-gw-1>enable
Password:
msk-hostel-ealkamal-gw-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-ealkamal-gw-1(config)#interface g0/1
msk-hostel-ealkamal-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-ealkamal-gw-1(config-if)#switchport mode trunk

msk-hostel-ealkamal-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-ealkamal-gw-1(config-if)#exit
msk-hostel-ealkamal-gw-1(config)#interface f0/1
msk-hostel-ealkamal-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-ealkamal-gw-1(config-if)#switchport mode trunk

msk-hostel-ealkamal-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-ealkamal-gw-1(config-if)#exit
msk-hostel-ealkamal-gw-1(config)#vlan 202
msk-hostel-ealkamal-gw-1(config-vlan)#name q42-management
msk-hostel-ealkamal-gw-1(config-vlan)#exit
msk-hostel-ealkamal-gw-1(config)#interface vlan202
msk-hostel-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

msk-hostel-ealkamal-gw-1(config-if)#no shutdown
msk-hostel-ealkamal-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-ealkamal-gw-1(config-if)#exit
msk-hostel-ealkamal-gw-1(config)#vlan 301
msk-hostel-ealkamal-gw-1(config-vlan)#name hostel-main
msk-hostel-ealkamal-gw-1(config-vlan)#exit
msk-hostel-ealkamal-gw-1(config)#interface vlan301
msk-hostel-ealkamal-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-ealkamal-gw-1(config-if)#no shutdown
msk-hostel-ealkamal-gw-1(config-if)#ip address 10.129.128.1 255.255.255.0
msk-hostel-ealkamal-gw-1(config-if)#exit
msk-hostel-ealkamal-gw-1(config)#exit
msk-hostel-ealkamal-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-ealkamal-gw-1#write memory
Building configuration...
[OK]
msk-hostel-ealkamal-gw-1#
```

```
sch-sochi-ealkamal-gw-1>enable
Password:
sch-sochi-ealkamal-gw-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-ealkamal-gw-1(config)#interface f0/0.401
sch-sochi-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

sch-sochi-ealkamal-gw-1(config-subif)#encapsulation dot1Q 401
sch-sochi-ealkamal-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-ealkamal-gw-1(config-subif)#description sochi-main
sch-sochi-ealkamal-gw-1(config-subif)#exit
sch-sochi-ealkamal-gw-1(config)#interface f0/0.402
sch-sochi-ealkamal-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

sch-sochi-ealkamal-gw-1(config-subif)#encapsulation dot1Q 402
sch-sochi-ealkamal-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-ealkamal-gw-1(config-subif)#description sochi management
sch-sochi-ealkamal-gw-1(config-subif)#description sochi-management
sch-sochi-ealkamal-gw-1(config-subif)#exit
sch-sochi-ealkamal-gw-1(config)#exit
sch-sochi-ealkamal-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-ealkamal-gw-1#write memory
Building configuration...
[OK]
sch-sochi-ealkamal-gw-1#
```

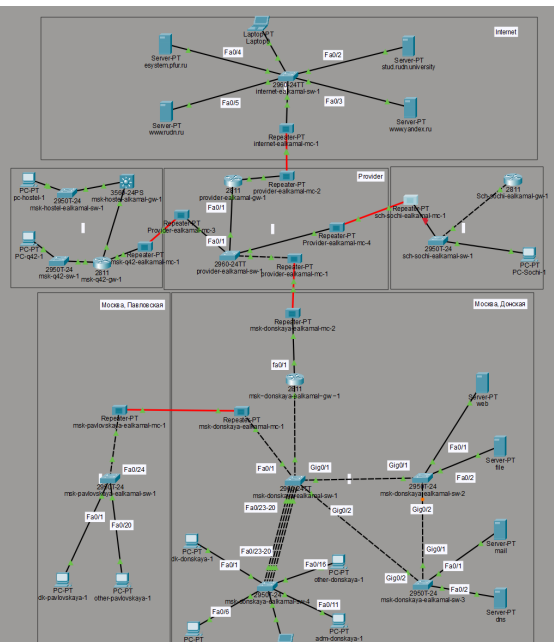
Рис. 9: Настройка VLAN sochi-main и sochi-management


```
[OK]
msk-donskaya-ealkamal-gw-1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-ealkamal-gw-1(config)#interface f0/1.5
msk-donskaya-ealkamal-gw-1(config-subif)#ip nat inside
msk-donskaya-ealkamal-gw-1(config-subif)#exit
msk-donskaya-ealkamal-gw-1(config)#interface f0/1.6
msk-donskaya-ealkamal-gw-1(config-subif)#ip nat inside
msk-donskaya-ealkamal-gw-1(config-subif)#exit
msk-donskaya-ealkamal-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#remark q42
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-ealkamal-gw-1(config-ext-nacl)#exit
msk-donskaya-ealkamal-gw-1(config)#
```

Рис. 16: Настройка NAT и ACL nat-inet

Итоговая топология сети

- Формирование итоговой топологии сети
- Подключение q42, hostel, Sochi и сети провайдера
- Интеграция модельной сети Интернет



Раздел 4

Выводы

- Настроены VLAN и trunk-соединения
- Реализована статическая маршрутизация между площадками
- Настроен NAT для взаимодействия с внешней сетью
- Построена итоговая топология сети с филиалами и Интернетом